The empirical recurrence for OEIS A302221, recovered from its tail

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Abstract

OEIS A302221 records "Empirical recurrence of order 92" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 160$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 98$. Its last coefficient is $c_{92} = -295944192 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302221 is "Number of n X 5 0..1 arrays with every element equal to 1, 2 or 3 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$3, 146, 760, 6110, 40395, 292781, 2043848, 14419536, \ \ldots$$

The entry states:

> Empirical recurrence of order 92 (see link above).

As of the "Last modified" line on the live entry (Oct 30 2022 15:40:55, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 160$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 160$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 92: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 92.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 320$ exact terms from $n = 6$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{92} c_i\, a(n-i),$$

c_{1}	6	c_{24}	137033758142	c_{47}	−265260722254446	c_{70}	392049674423185
c_{2}	33	c_{25}	4457102058	c_{48}	184561538083608	c_{71}	8831418323296
c_{3}	−91	c_{26}	−373258195269	c_{49}	464689202008341	c_{72}	−160442268799128
c_{4}	−784	c_{27}	−572947937813	c_{50}	80461586577375	c_{73}	−172055539927093
c_{5}	−640	c_{28}	62847340039	c_{51}	−464069301620712	c_{74}	3510673170529
c_{6}	9217	c_{29}	1410388480324	c_{52}	−292015951204005	c_{75}	36108544973443
c_{7}	29846	c_{30}	1880546595021	c_{53}	369297321057874	c_{76}	69629720657430
c_{8}	−27804	c_{31}	−176386061731	c_{54}	639910513907695	c_{77}	3666992209710
c_{9}	−246238	c_{32}	−3561620949580	c_{55}	−244573214073668	c_{78}	−3892668114204
c_{10}	−680659	c_{33}	−4219778845480	c_{56}	−875652938966921	c_{79}	−24695906397232
c_{11}	−342576	c_{34}	−620453692494	c_{57}	−442510989026347	c_{80}	−3350768213206
c_{12}	6910404	c_{35}	2851802450021	c_{58}	900677256431667	c_{81}	728115443632
c_{13}	24072843	c_{36}	3883543961444	c_{59}	1100639112762413	c_{82}	6459800075496
c_{14}	−508980	c_{37}	9014672256092	c_{60}	−83933388580096	c_{83}	1080045490920
c_{15}	−192686988	c_{38}	21083631101147	c_{61}	−1265868589210777	c_{84}	−405032829040
c_{16}	−379730411	c_{39}	7723612306659	c_{62}	−898161843753887	c_{85}	−1022586018784
c_{17}	416835612	c_{40}	−52341769831874	c_{63}	542622194188761	c_{86}	−175603127104
c_{18}	2842736399	c_{41}	−100637776706247	c_{64}	1154893304703333	c_{87}	86065157120
c_{19}	2826739294	c_{42}	−20756846468196	c_{65}	472650307070876	c_{88}	74284301824
c_{20}	−7724541022	c_{43}	170952919403012	c_{66}	−621719532152082	c_{89}	13545382912
c_{21}	−24347289505	c_{44}	219632266152334	c_{67}	−762575872485203	c_{90}	−5782960128
c_{22}	−9572720010	c_{45}	−17527846547342	c_{68}	−138560043032012	c_{91}	−954408960
c_{23}	68508299697	c_{46}	−353015944636795	c_{69}	403407023223281	c_{92}	−295944192

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \ge 98$, and it is the recurrence OEIS A302221 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1 x^{92-1} - \cdots - c_{92}$. Its constant term is $-c_{92} = 295944192$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 92, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -295944192.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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